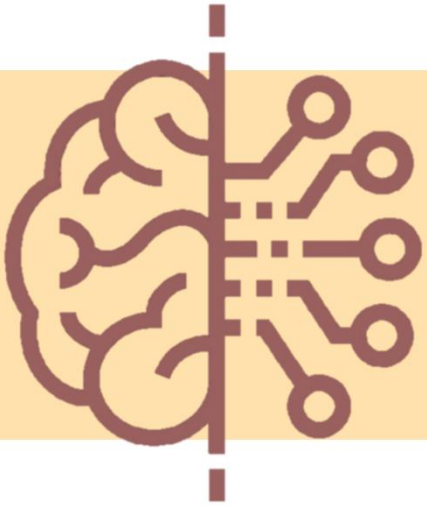
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Machine Learning



Artificial Intelligence

School of Computing
Universiti Teknologi Malaysia

Premier Digital Tech University ™

innovative ● entrepreneurial ● global

Machine Learning

How do we teach a machine to learn?

Taxonomy

- Learning types
  - Supervised learning
  - Unsupervised learning
  - Reinforcement learning
- Machine learning process
- From data to features
- Machine learning techniques
  - Clustering, classification, regression
- Evaluation metrics
- Computational intelligence

Machine Learning

Learning Types

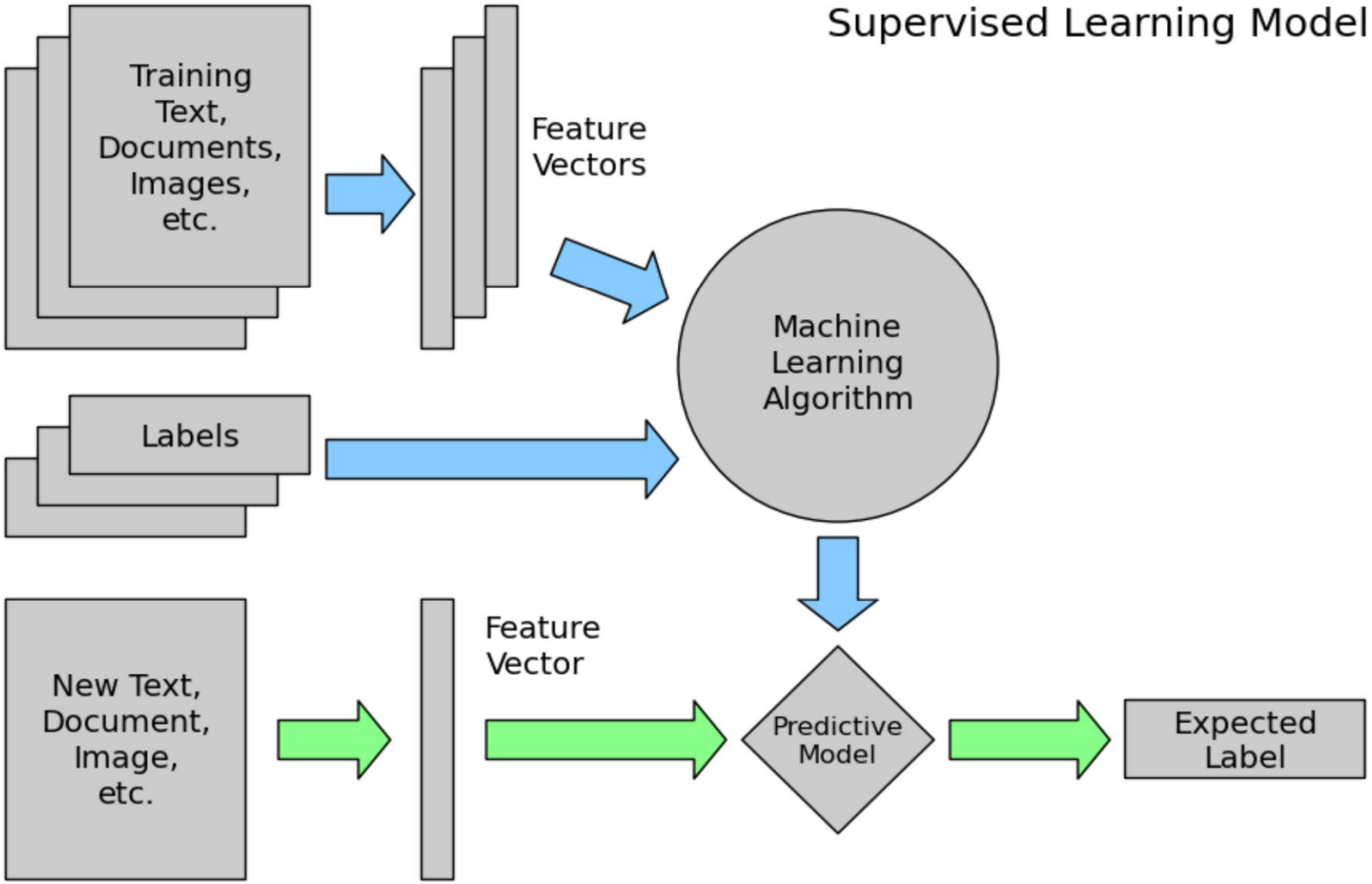
Learning types

- Supervised learning
  - The correct classes of the training data are known
- Unsupervised learning
  - The correct classes of the training data are NOT known
- Reinforcement learning
  - Machine learn its behavior based on feedback from environment

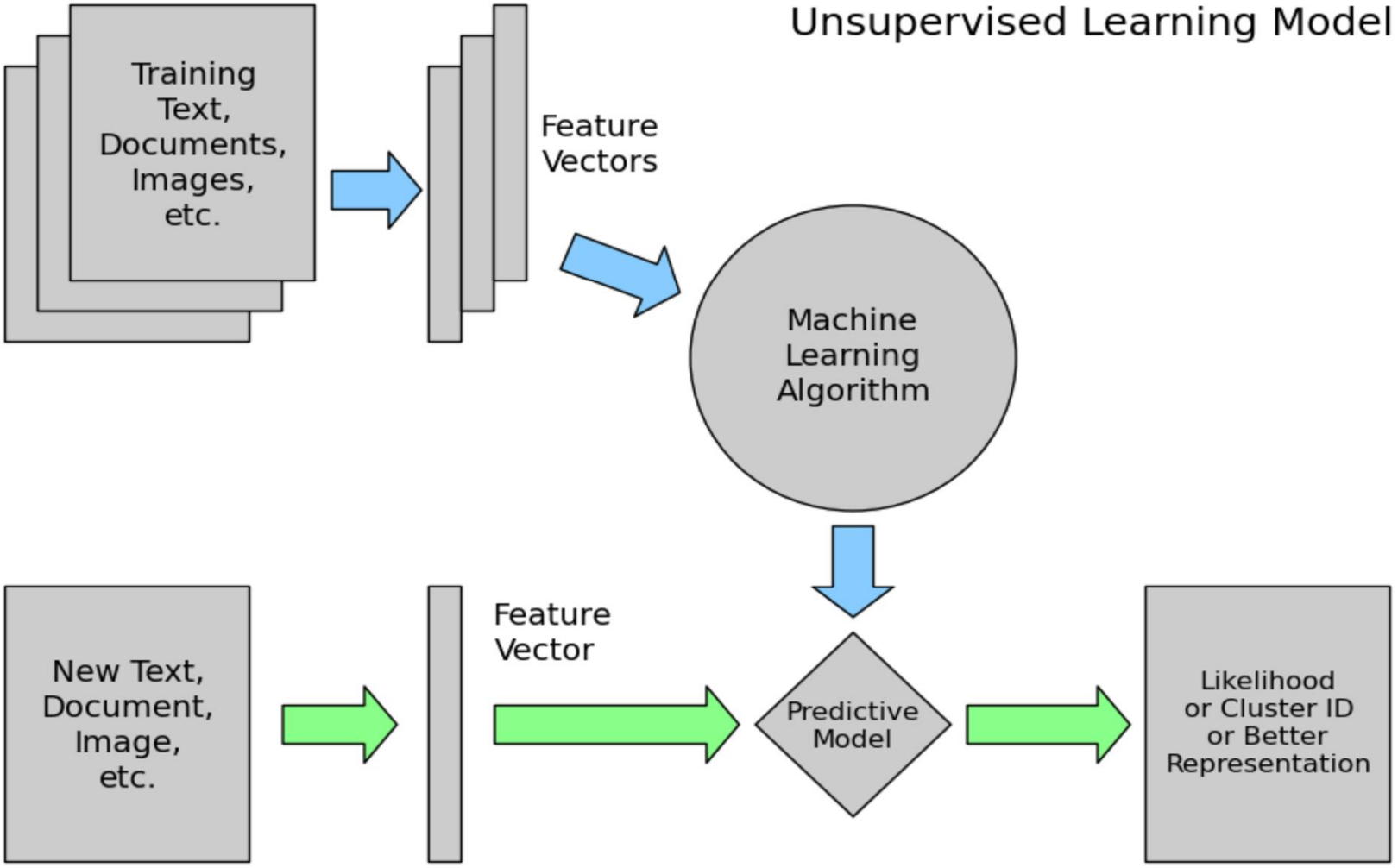
Learning types

	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	The machine learns by using labelled data	The machine is trained on unlabelled data without any guidance	An agent interacts with its environment by producing actions & discovers errors or rewards
Type of problems	Regression & Classification	Association & Clustering	Reward based
Type of data	Labelled data	Unlabelled data	No pre-defined data
Training	External supervision	No supervision	No supervision
Approach	Map labelled input to known output	Understand patterns and discover output	Follow trail and error method
Popular algorithms	Linear regression, Logistic regression, Support Vector Machine, KNN, etc	K-means, C-means, etc	Q-Learning, SARSA, etc

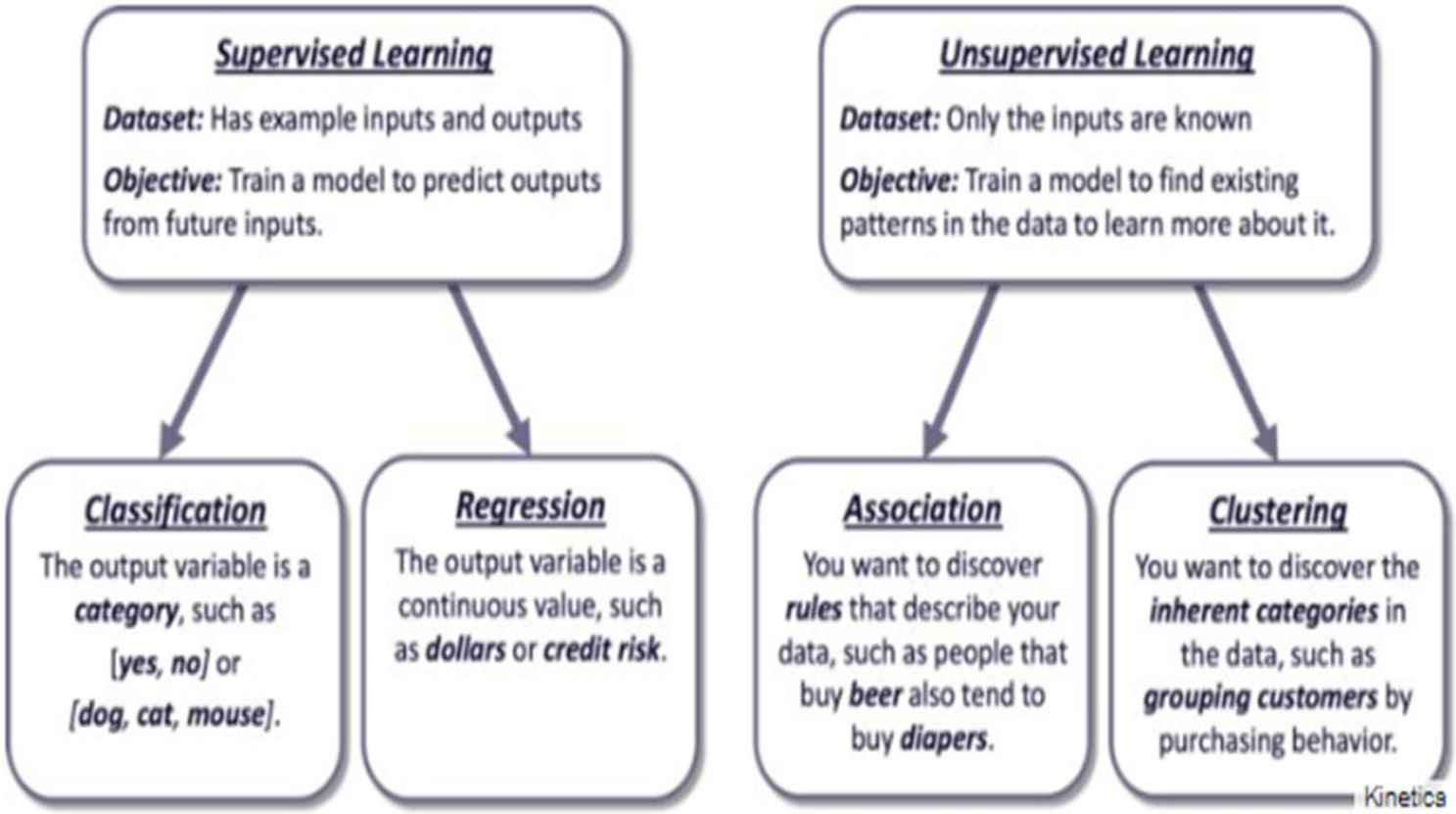
1. Supervised Learning



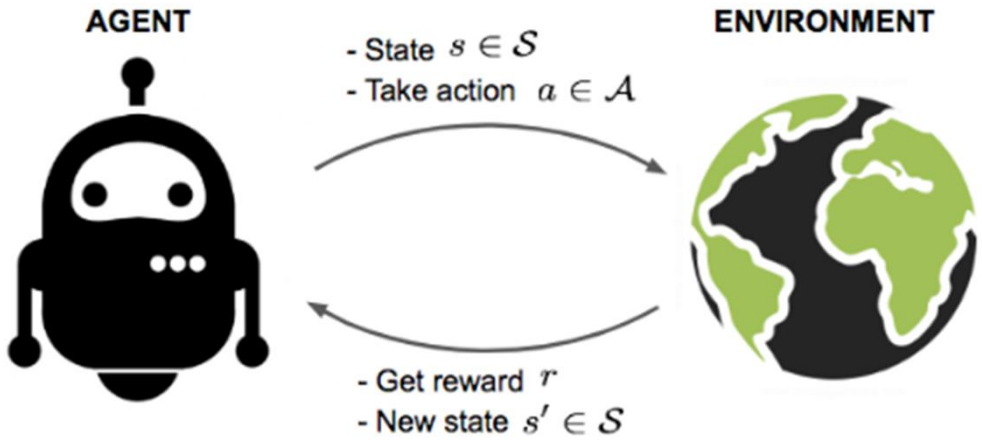
2. Unsupervised Learning



Supervised vs unsupervised

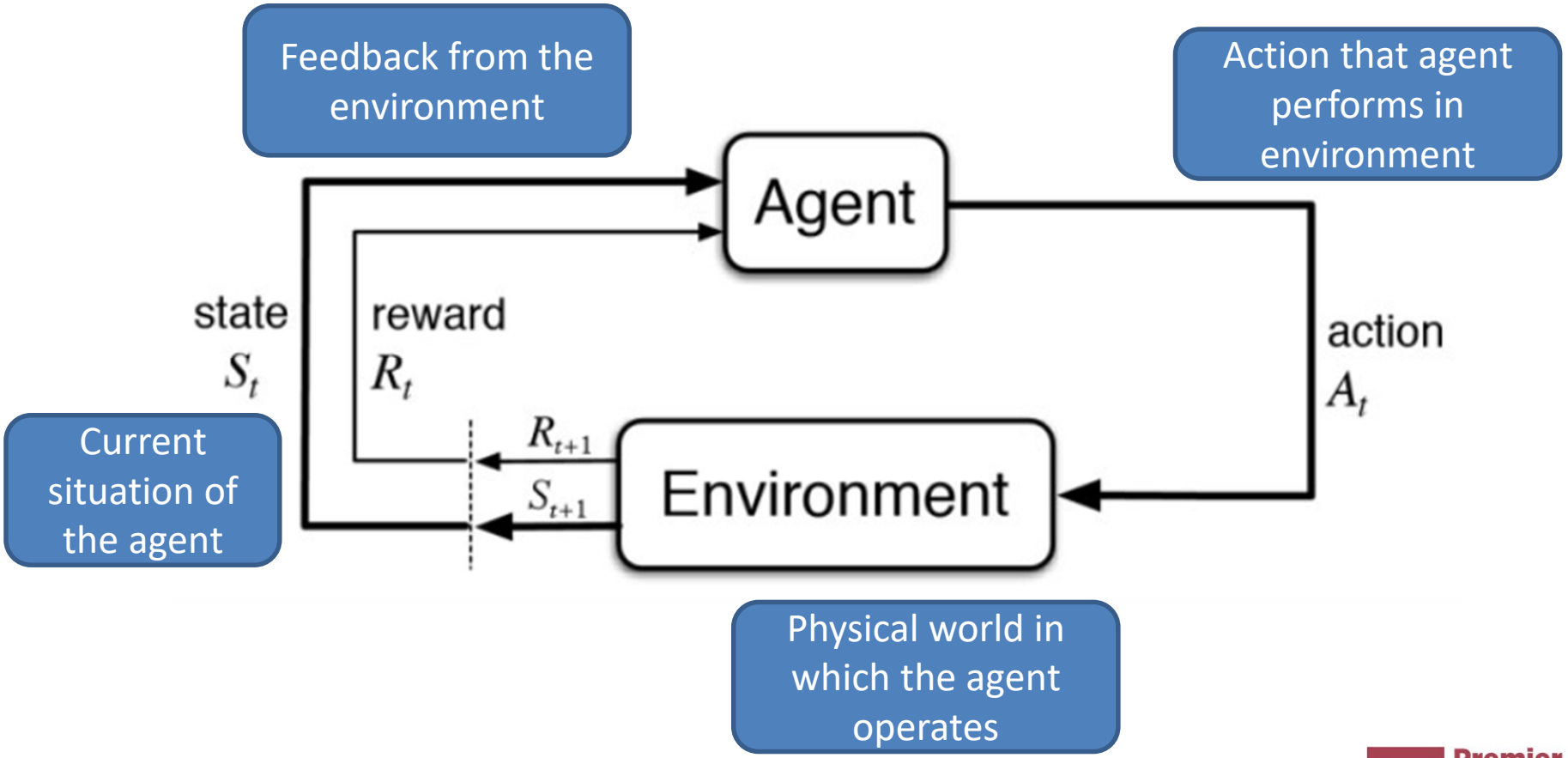


3. Reinforcement learning



- An agent interacts with the environment trying to take smart actions to maximize cumulative rewards
- The environment is formulated using Markov decision process to search for solutions

Markov Decision Process (MDP)



Markov Decision Process (MDP)

- The agent is acting in an **environment**
- How the environment reacts to certain actions is defined by a **model** which we may or may not know
- The agent can stay in one of the many **states** S of the environment and choose to take one of the many **actions** A so that it can switch from one state to another
- Which state the agent will arrive is determined by transition **probabilities** P between states
- Once an action is taken, a **reward** R is delivered by the environment as feedback

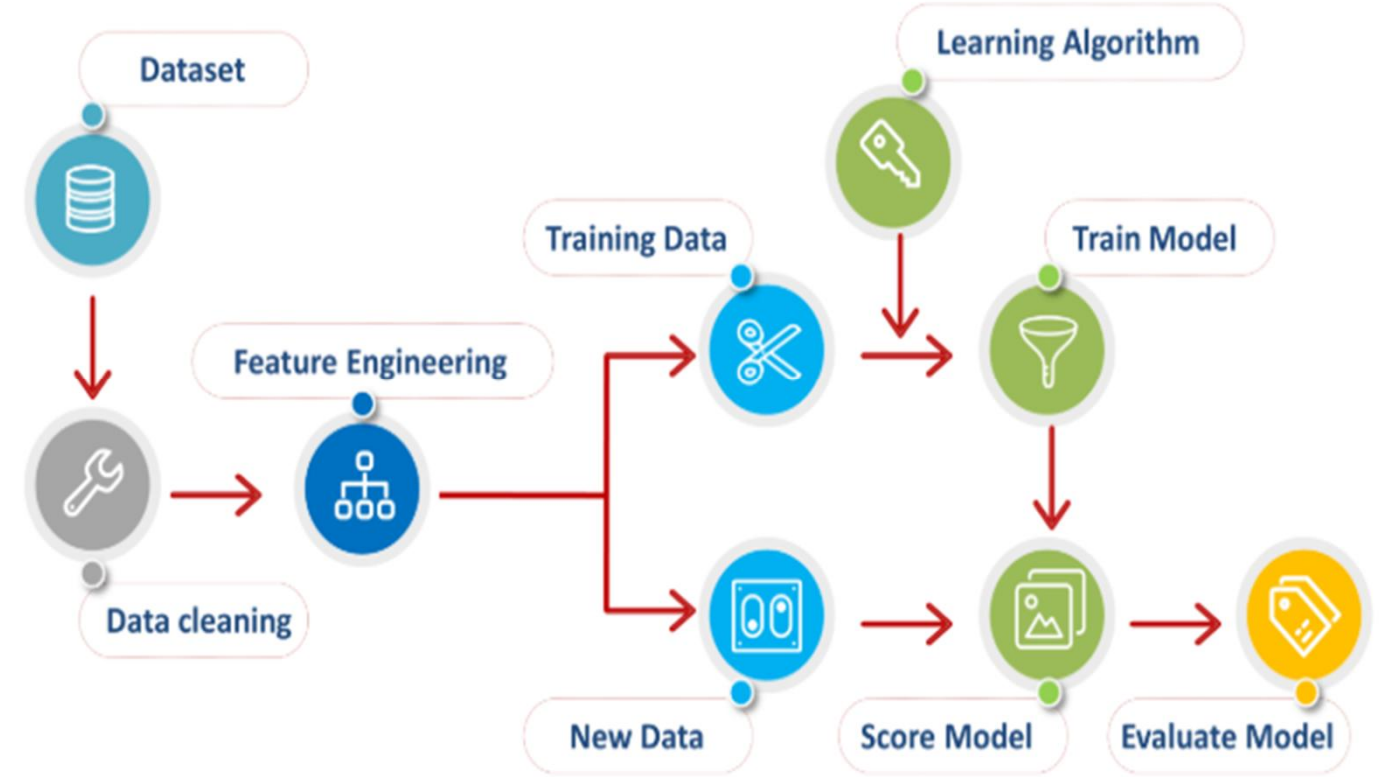
RL in real world

- Games
  - AlphaGo, AlphaGo Zero etc
- Robotic
  - (remember Spot from BostonDynamics?)
- Traffic light control
- Web system configuration
- And many more

Machine Learning

Learning Process

Machine Learning Process



How machine learns?

- Machine learning utilizes techniques of understanding **patterns** from input **data**
- Data is used for **training** by learning algorithm to **learn** and develop a specific computational **model**
- **Pre-processing** involves extracting important features of data
- **Model** refers to predefined prototype problem solver, produced based on patterns of input data

Machine Learning

From Data to Features

Some real data

- *In the 1920's, botanists collected measurements on the*
  1) sepal length
  2) sepal width
  3) petal length
  4) petal width



*of 150 iris, 50 from each of three species (*setosa, versicolor, virginica)
The measurements became known as ***Fisher's iris data***







Iris data

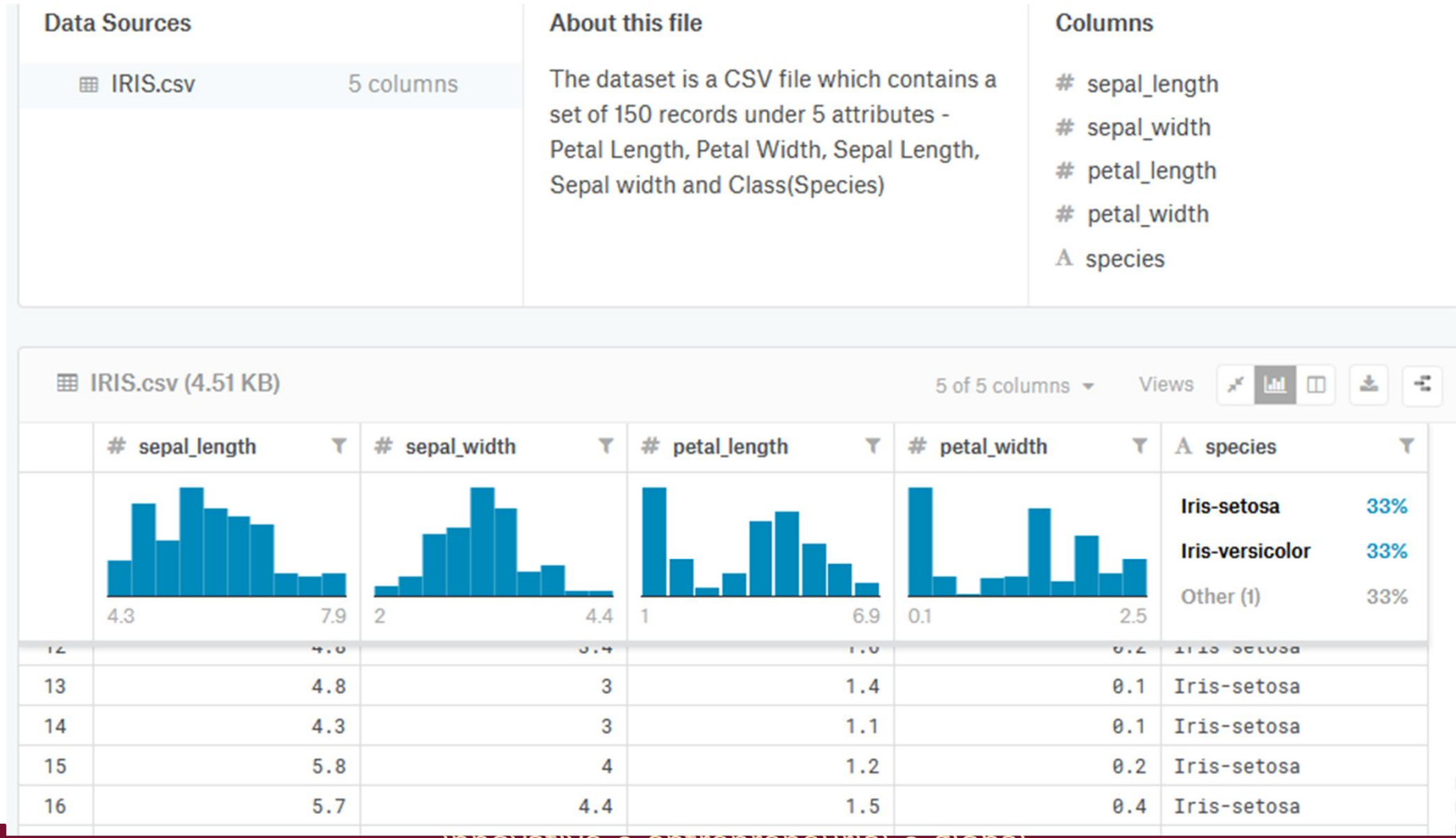
```
>> load fisheriris
>> size(meas)
   150     4
>> meas
```

	Feature #1	Feature #2	Feature #3	Feature #4
Sample #1	6.3000	2.8000	5.1000	1.5000
Sample #2	6.1000	2.6000	5.6000	1.4000
Sample #3	7.7000	3.0000	6.1000	2.3000
	6.3000	3.4000	5.6000	2.4000
	6.4000	3.1000	5.5000	1.8000
	6.0000	3.0000	4.8000	1.8000
Sample #n	6.9000	3.1000	5.4000	2.1000

```
etc...
>> size(species)
   150     1
>> species
'versicolor' ,'versicolor', 'virginica', 'virginica', 'virginica',
   'setosa', 'setosa', 'setosa', 'setosa', 'setosa', 'setosa',  etc...
```

Iris data

- https://www.kaggle.com/arshid/iris-flower-dataset



Data Sources

IRIS.csv 5 columns

About this file

The dataset is a CSV file which contains a set of 150 records under 5 attributes - Petal Length, Petal Width, Sepal Length, Sepal width and Class(Species)

Columns

sepal_length
sepal_width
petal_length
petal_width
A species

IRIS.csv (4.51 KB) 5 of 5 columns Views

	# sepal_length	# sepal_width	# petal_length	# petal_width	A species
	4.3 – 7.9	2 – 4.4	1 – 6.9	0.1 – 2.5	Iris-setosa 33%, Iris-versicolor 33%, Other (1) 33%
13	4.8	3	1.4	0.1	Iris-setosa
14	4.3	3	1.1	0.1	Iris-setosa
15	5.8	4	1.2	0.2	Iris-setosa
16	5.7	4.4	1.5	0.4	Iris-setosa

Data to Features

- Collection of data observations and their attributes/features

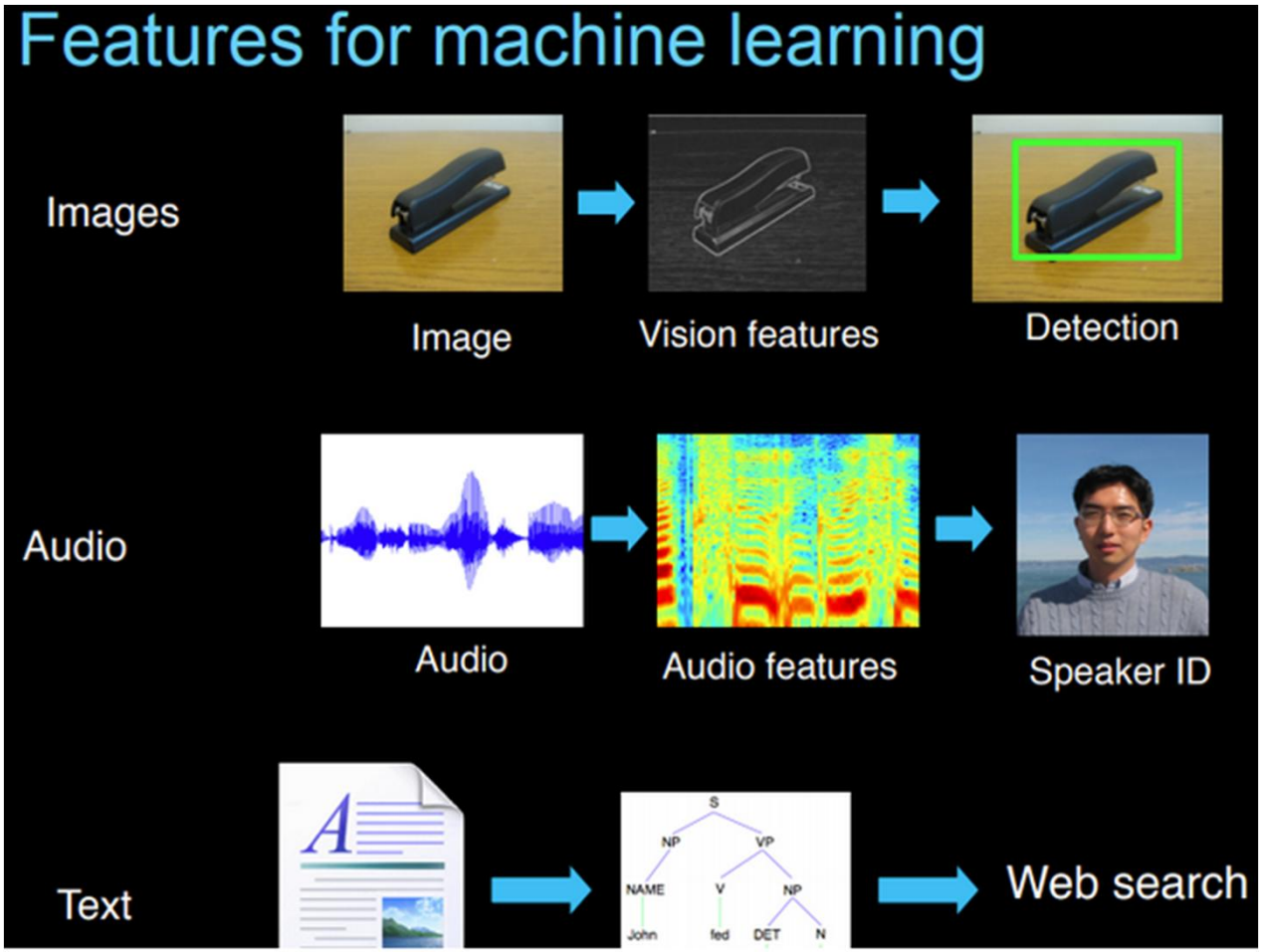
▪Attribute is the properties or characteristics of the data object
▪ Attribute is aka feature / field / characteristic / variable

1	id	age	sex	region	income	married	children	car	save_act	current_act	mortgage	pep
2	ID12101	48	FEMALE	INNER_CITY	17546	NO	1	NO	NO	NO	NO	YES
3	ID12102	40	MALE	TOWN	30085.1	YES	3	YES	NO	YES	YES	NO
4	ID12103	51	FEMALE	INNER_CITY	16575.4	YES	0	YES	YES	YES	NO	NO
5	ID12104	23	FEMALE	TOWN	20375.4	YES	3	NO	NO	YES	NO	NO
6	ID12105	57	FEMALE	RURAL	50576.3	YES	0	NO	YES	NO	NO	NO
7	ID12106	57	FEMALE	TOWN	37869.6	YES	2	NO	YES	YES	NO	YES
8	ID12107	22	MALE	RURAL	8877.07	NO	0	NO	NO	YES	NO	YES
9	ID12108	58	MALE	TOWN	24946.6	YES	0	YES	YES	YES	NO	NO
10	ID12109	37	FEMALE	SUBURBAN	25304.3	YES	2	YES	NO	NO	NO	NO

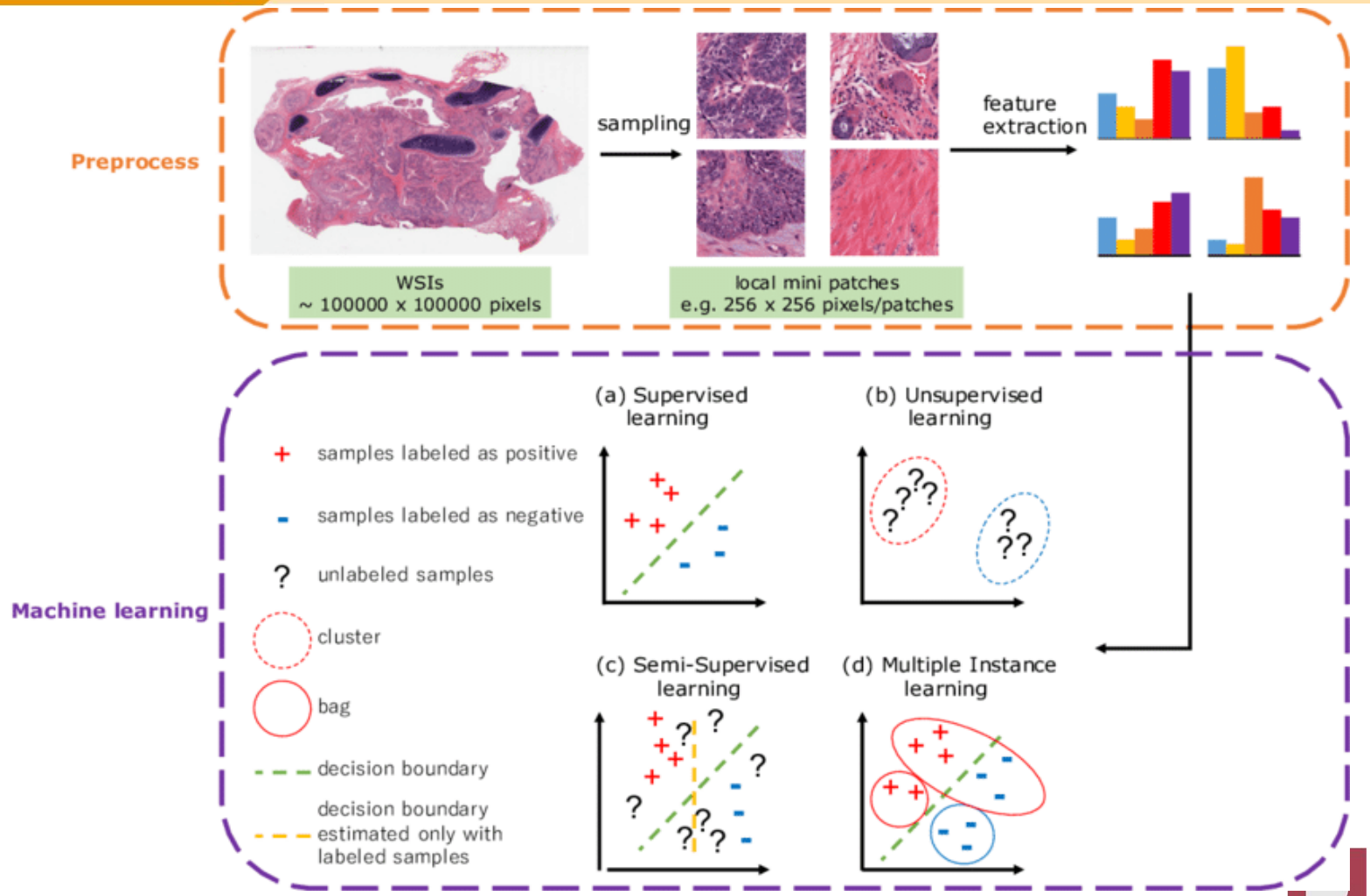
Observation/instances/examples/samples/objects/records

Class

Features can be from many forms of data; images, audio or text



Real world example: Steps of ML in digital pathological image analysis



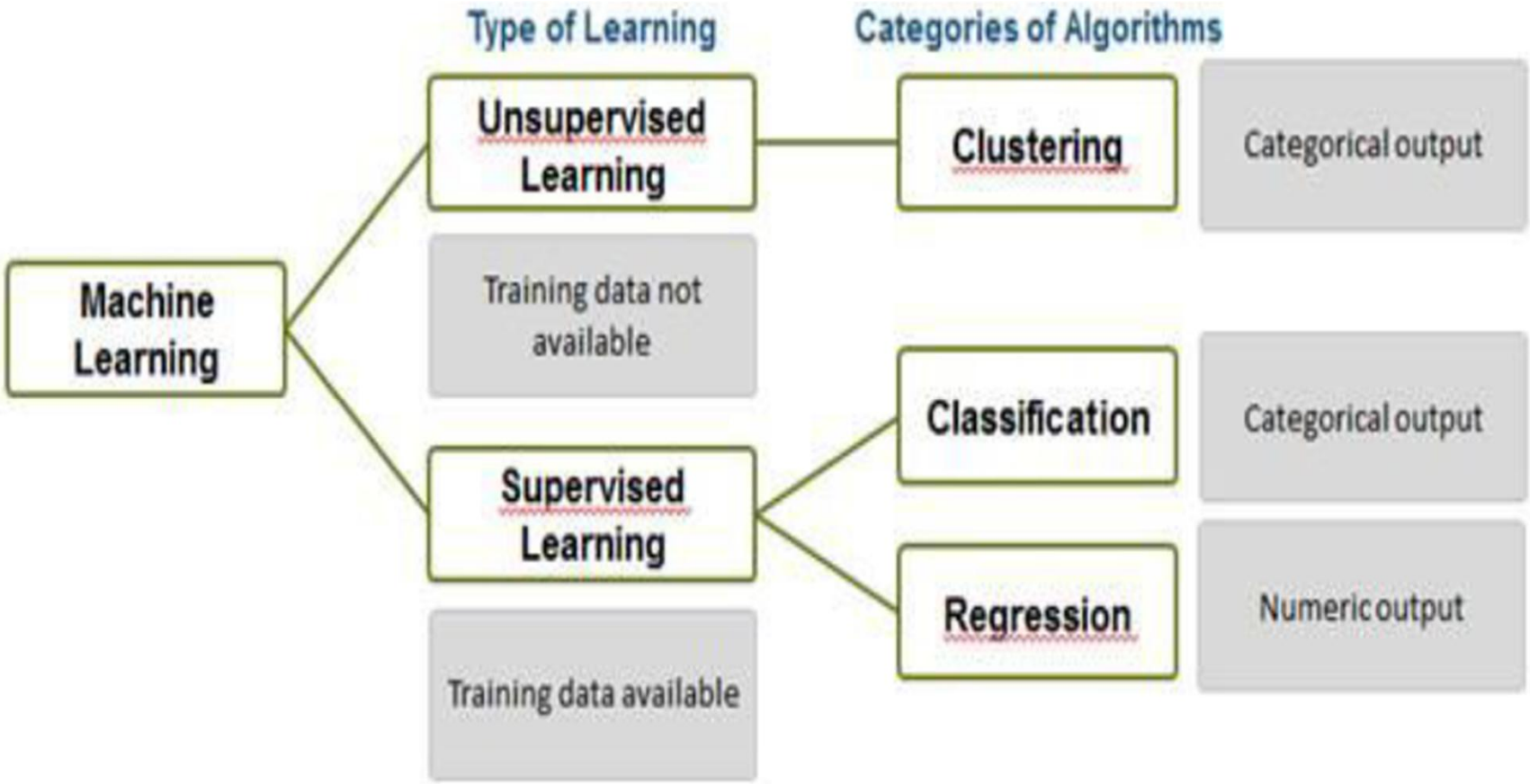
Machine Learning

Learning Techniques

Machine learning techniques

- Classification
  - Predict class from observation
- Clustering
  - Group observations into ‘’meaningful” groups
- Regression
  - Predict value from observations

Classification and clustering



Classification

- Classification refers to predicting the class of given observations/data points
- Class means targets/labels or sometimes categories
- In classification, we are mapping the input data to specific type of class/category

Classification



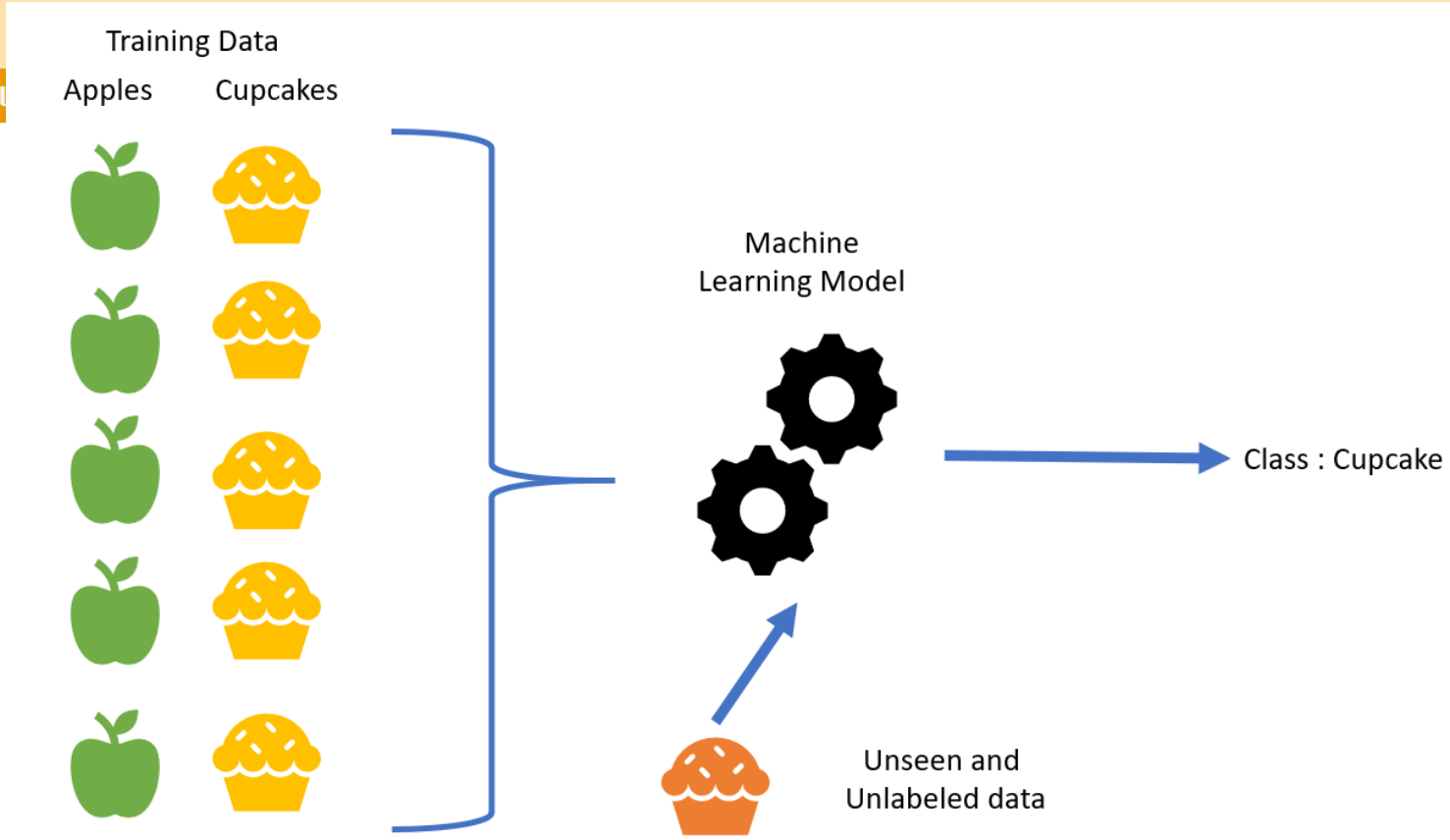
- Classification of waste
- You know beforehand each bin category/class of the waste
- How do you decide which waste goes to which bin?
- We learn from the waste i.e. from the data points, then we can decide to which class it belongs to

Classification



- The data that gets input to the classifier may contains four measurements for waste e.g. 1)what material it is made from, 2)size, 3)weight and its 4)biodegradable level
- The job of the classifier then is to output the correct waste type for every input

How classification works



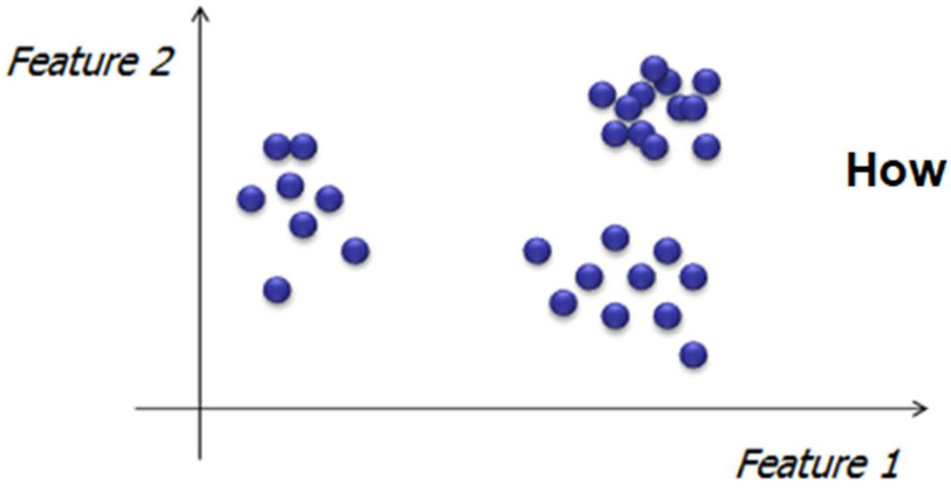
- Data set = features and class label
- Data separated int training data and testing data
- ML algorithm predicts the class label of the testing data

Classification algorithms

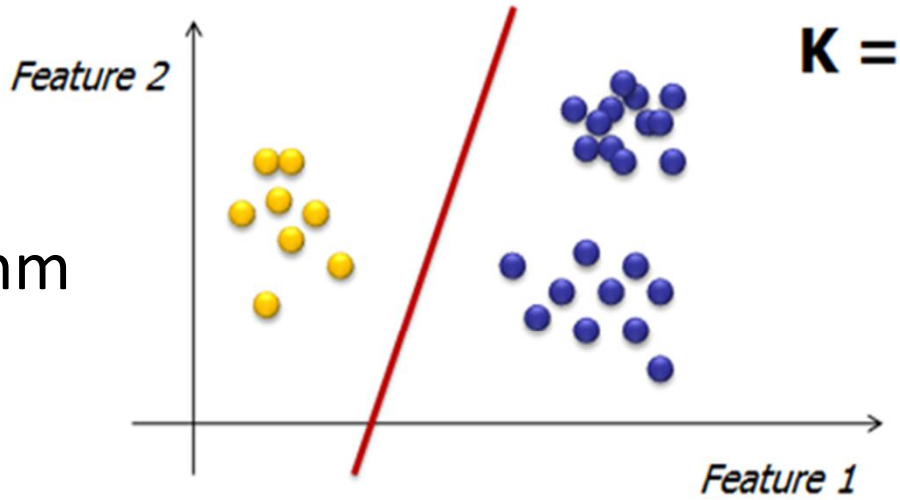
- Decision tree
- Naïve Bayes
- ANN, now evolving into Deep Learning
- Random Forest
- K-Nearest Neighbour
- Support Vector Machine (SVM)
- And many more!

Clustering

- Clustering is identifying similar groups of data in a dataset
- We assume that observations in each group are comparatively more similar to observations of that group compared to those of the other groups
- A cluster refers to a collection of data points aggregated together because of certain similarities
- E.g. try to understand customers better so that we can market our products to them better. How do we start?
- E.g. try to find your close-knit groups of friends on Facebook? How will you identify them?

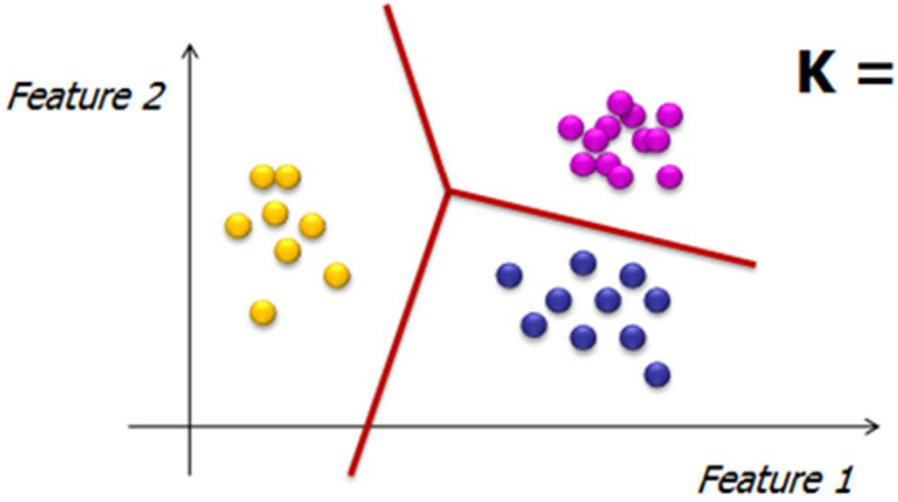


How many cluster?



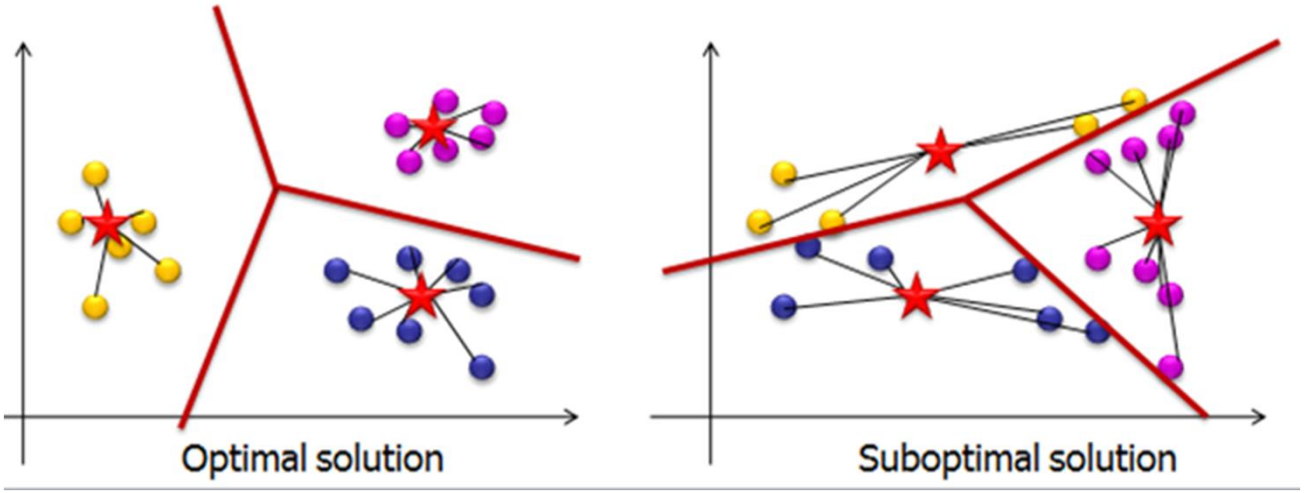
K = 2

Clustering using k-mean algorithm



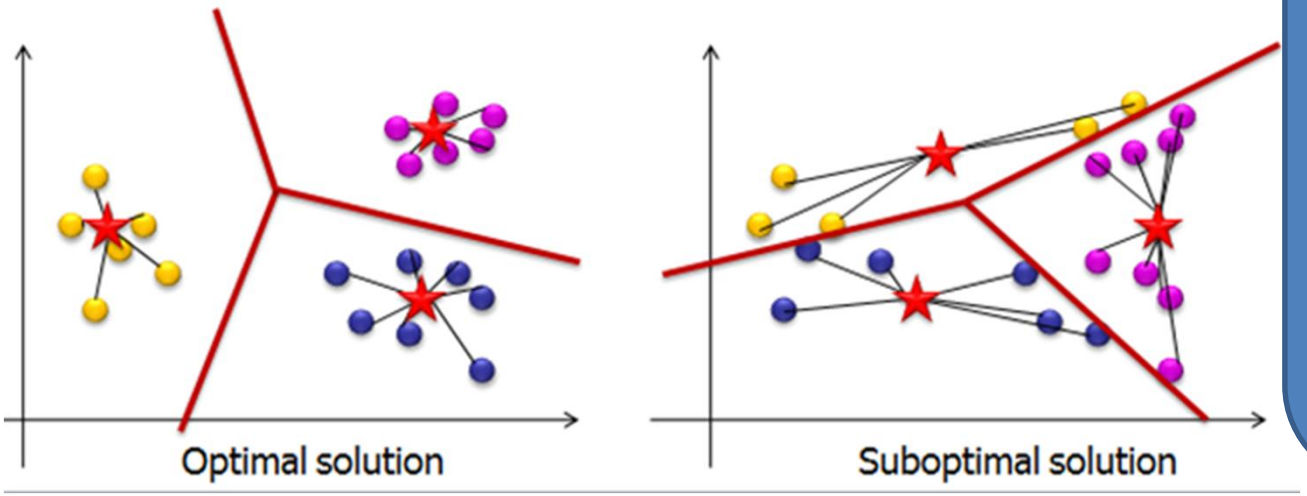
K = 3

Clustering (*k*-mean algorithm)



- Find similarities among all observations such that similar observations can be put in clusters
- Computing distance from each observation/data point to centroids to obtain well-represented clusters.
- Why are they optimal and suboptimal solution? Can you guess?

Clustering (*k*-mean algorithm)



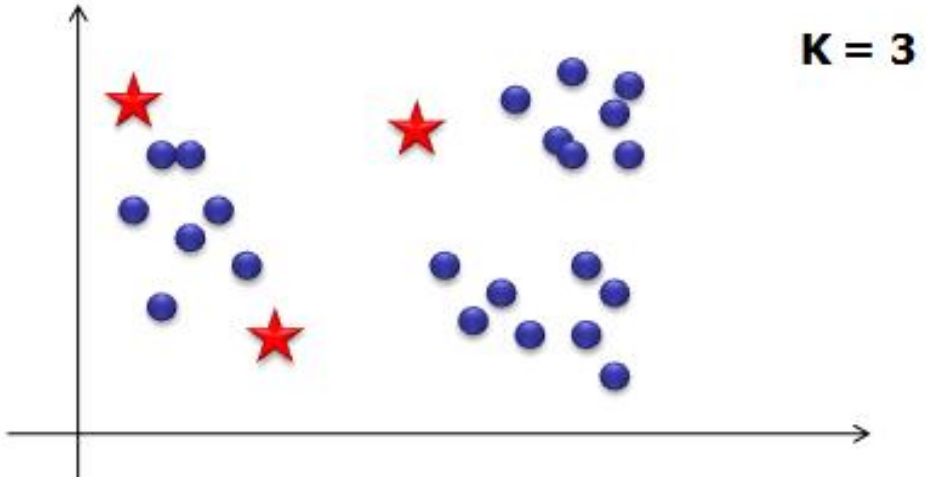
Centroid == mean i.e. averaging the data

Data points are allocated to the nearest centroid/cluster as to keep the centroids as small as possible

- Find similarities among all observations such that similar observations can be put in clusters
- Computing distance from each observation/data point to the centroids to obtain well-represented clusters.
- Why are they optimal and suboptimal solution? Can you guess?

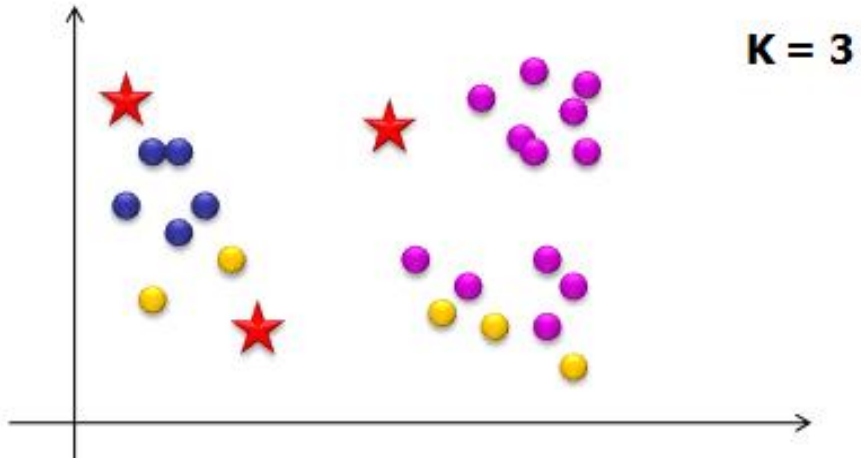
Clustering (*k*-mean algorithm)

- **Algorithm description**
  - Choose the number of clusters, K
  - Randomly choose initial positions of K centroids
  - Assign each of the points to the "nearest centroid" (depends on distance measure)



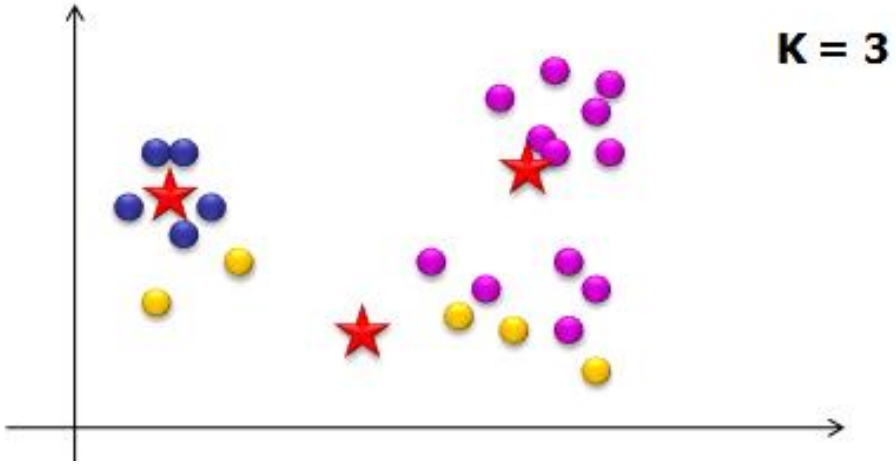
Clustering (*k*-mean algorithm)

- **Algorithm description**
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  - Assign each of the points to the "nearest centroid" (depends on distance measure)



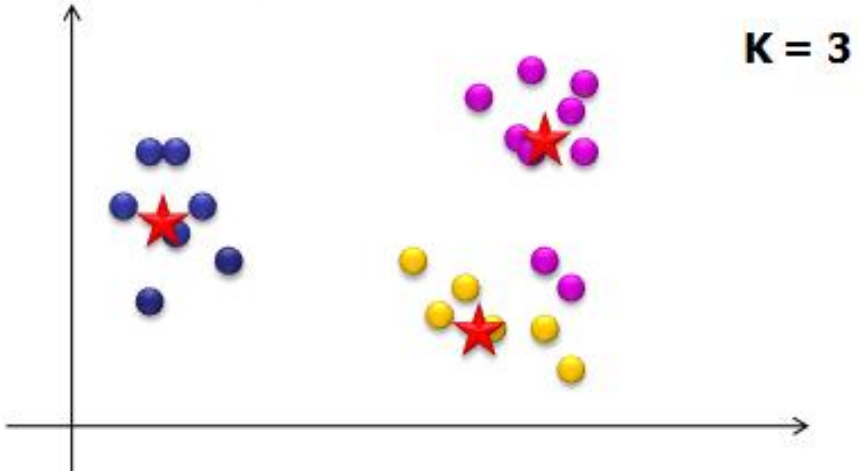
Clustering (*k*-mean algorithm)

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  - Choose the number of clusters - K
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  - Assign each of the points to the "nearest centroid" (depends on distance measure)
  - Re-compute centroid positions
  - If solution converges → Stop!



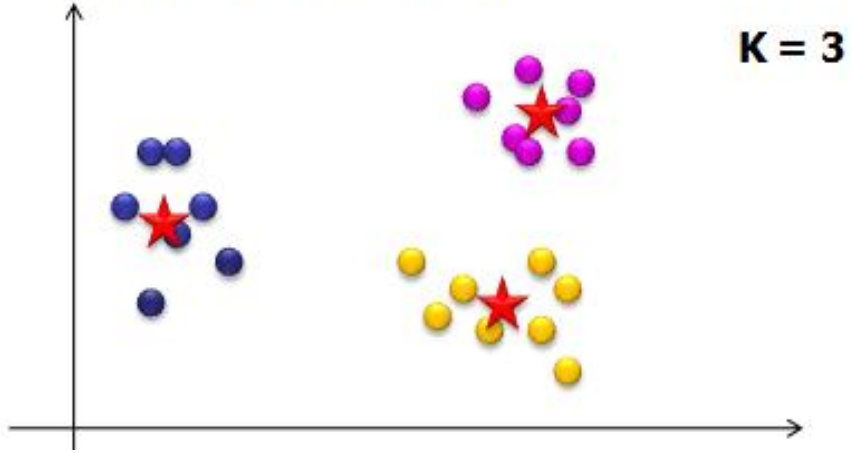
Clustering (*k*-mean algorithm)

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Clustering (*k*-mean algorithm)

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  - Choose the number of clusters - K
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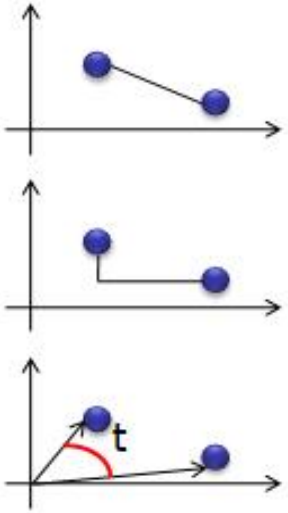


Distance measure for *k*-mean

- How should we choose K?
- What type of distance measures can we use, and how to choose between them?
  - Euclidean

    $((x_2 - x_1)^2 + (y_2 - y_1)^2)^{0.5}$
  - Sum of absolute differences

    $|x_2 - x_1| + |y_2 - y_1|$
  - 1 – Cos(t)
  - And more...



Machine Learning

Performance Evaluation

How good is the learning model?

- The aim of a model is to be able to produce good results
- Employ another set of data called **testing data** to **evaluate** the learning **model**
  - Can be completely new
  - Blinded dataset from the same resource
  - Smaller part of training data previously separated
  - Rule of thumb for train-test split (holdout method); 70%:30, 80%:20%, 90%:10%

Performance measures / Evaluation metrics

- Confusion matrix to evaluate the performance of ML methods

		Predicted Class	
		Yes	No
Actual Class	Yes	True Positive (TP)	False Negative (FN)
	No	False Positive (FP)	True Negative (TN)

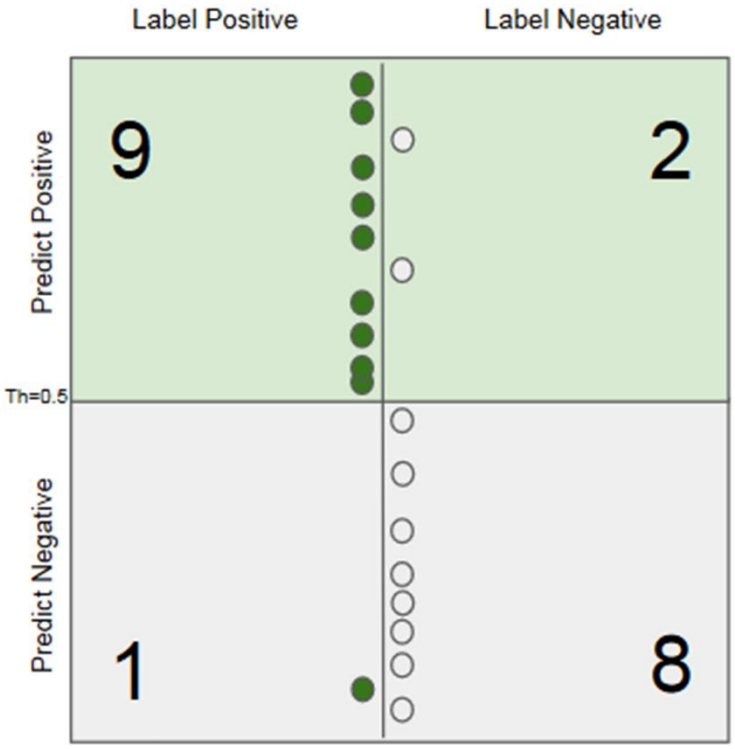
- E.g. of performance measures
  - % game won
  - % correctly classified images
  - % correctly identified text

Confusion matrix



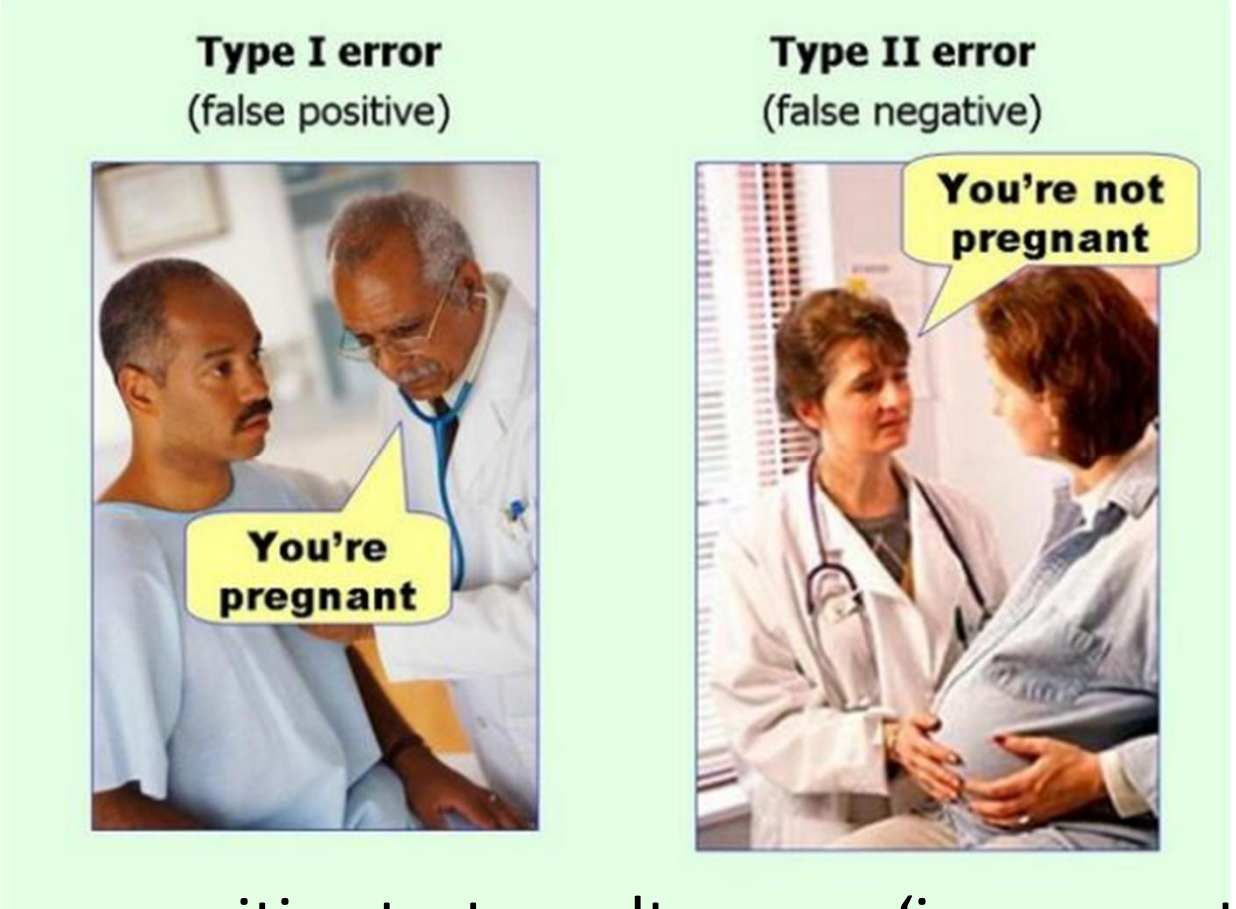
●	Positive labelled example
○	Negative labelled example

		Predicted Class	
		Yes	No
Actual Class	Yes	True Positive (TP)	False Negative (FN)
	No	False Positive (FP)	True Negative (TN)



- TP = 9 (positive observation correctly identified as positive)
- TN = 8 (negative observation correctly identified as negative)
- FP = 2 (positive that is *false* i.e. negative observation incorrectly identified as positive)
- FN = 1 (negative that is *false* i.e. positive observation incorrectly identified as negative)

Type I error = FP and Type II error = FN



- Lets say positive test result means 'is pregnant'

Performance measures from confusion matrix

		Predicted Class	
		Yes	No
Actual Class	Yes	True Positive (TP)	False Negative (FN)
	No	False Positive (FP)	True Negative (TN)

- Accuracy = TP+TN / (TP+TN+FP+FN) i.e. 1 – error
- Precision = TP/(TP+FN)
- Recall = TP/(TP+FP)
- The closer the score to 1, the better the classifier model
- **High recall, low precision:** This means that most of the positive examples are correctly recognized (low FN) but there are a lot of false positives.
- **Low recall, high precision:** This shows that we miss a lot of positive examples (high FN) but those we predict as positive are indeed positive (low FP)

Activity 1: Confusion matrix

Sepal length	Sepal width	Petal length	Petal width	Actual Class	Predicted Class
4.4	3.2	1.3	0.2	1	1
5	3.5	1.6	0.6	1	0
5.1	3.8	1.9	0.4	1	0
4.8	3	1.4	0.3	0	0
5.1	3.8	1.6	0.2	0	0
4.6	3.2	1.4	0.2	0	0
5.3	3.7	1.5	0.2	1	1
5	3.3	1.4	0.2	1	1
7	3.2	4.7	1.4	0	0
6.4	3.2	4.5	1.5	1	1
6.9	3.1	4.9	1.5	1	1
5.5	2.3	4	1.3	0	1
6.5	2.8	4.6	1.5	0	1
5.7	2.8	4.5	1.3	0	0
6.3	3.3	4.7	1.6	0	0
4.9	2.4	3.3	1	1	1
6.6	2.9	4.6	1.3	1	1
5.2	2.7	3.9	1.4	0	1
5	2	3.5	1.4	0	0
5.1	2.3	3.6	1.2	0	1

- Let positive be 1, and negative is 0;
- Given the Iris dataset as in the table, quantify TP, TN, FP and FN
- Determine the accuracy, precision and recall

Activity 1 Answer: Confusion matrix

	Predicted: 0	Predicted: 1
Actual: 0	7	2
Actual: 1	4	7

- **TP=7, TN=7, FN=2, FP=4**
- Accuracy = TP+TN / (TP+TN+FP+FN)

  =7+7/ (7+7+2+4) = 0.7
- Precision = TP/(TP+FN) = 7/(7+2)=0.78
- Recall = TP/(TP+FP)= 7/(7+4) = 0.64

How real world use ML?

https://towardsdatascience.com/real-world-examples-of-applied-machine-learning-from-ai-conference-4d4678700c6

Real world examples

- Uber COTA (Customer Obsession Ticket Assistant)

http://w4nderlu.st/assets/javascript/ViewerJS/#http://w4nderlu.st/content/3-publications/cota-improving-the-speed-and-accuracy-of-customer-support-through-ranking-and-deep-networks/cota-o-reilly-ai-conference-2018.pdf

- AirBNB ; categorizing listing photos

https://medium.com/airbnb-engineering/categorizing-listing-photos-at-airbnb-f9483f3ab7e3

- ZocDoc ; finding in network physicians from insurance cards

https://www.zocdoc.com/about/blog/tech/making-sense-of-insurance-cards-using-deep-learning/

- Spotify ; How does it know your kind of music?

https://medium.com/s/story/spotifys-discover-weekly-how-machine-learning-finds-your-new-music-19a41ab76efe

Computational Intelligence

What methods machines use to learn?

Outline

- What is AI?
- Now, come Computational Intelligence (CI)
- Scope of CI
- Importance of CI
- CI Techniques
  - Artificial Neural Network
  - Genetic Algorithm
  - Fuzzy Logic

What is AI?

- Ability to interact with the real world
- Reasoning and Planning
- Learning and Adaptation



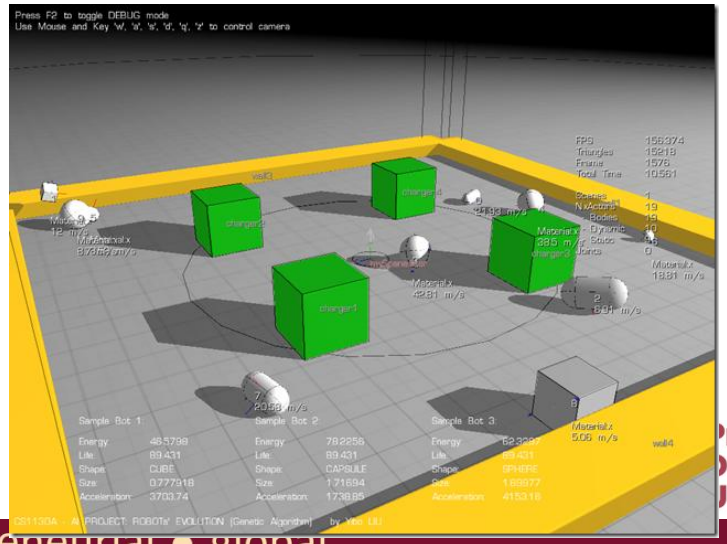
Now, come Computational Intelligence (CI)

- A methodology involving computing that exhibits an ability to learn and/or to deal with new situations, such that the system is perceived to possess one or more attributes of reason, such as generalization, discovery, association and abstraction (Russel et al, 1998).
- CI is the field of computing that draws from the successes of natural systems to develop new ways of solving computational problems in the real world.

Scope of CI

- Practical adaptation concepts, paradigms, algorithms and implementations that enable or facilitate appropriate actions (intelligent behavior) in complex and changing environments.



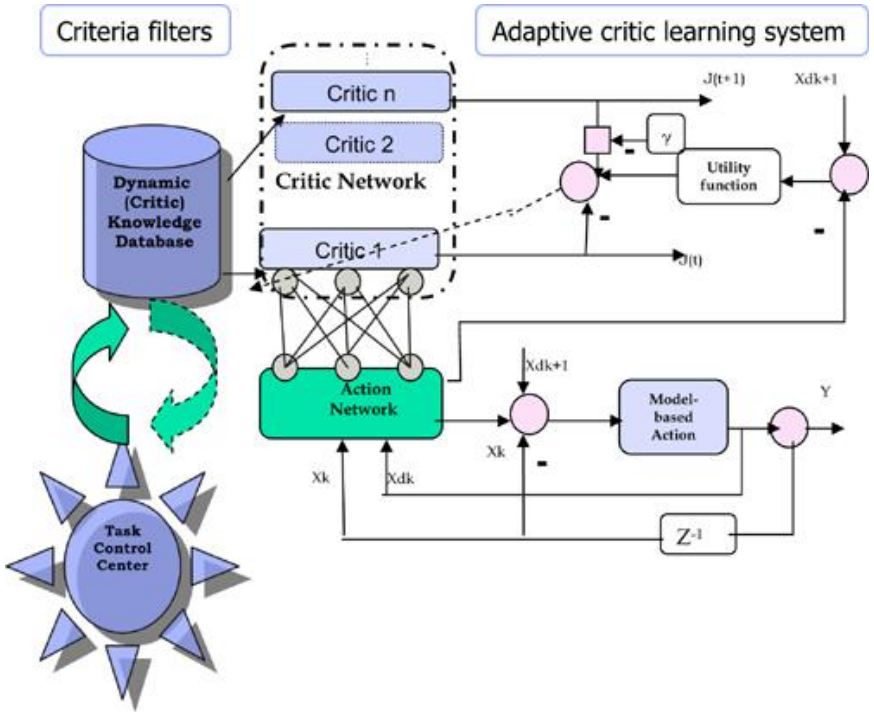


Scope of CI

- Fuzzy, imprecise or imperfect data
- No available mathematical algorithm
- Optimal solution unknown
- Rapid prototyping required
- Only domain experts available
- Robust system required
- Learning capability

More recently:

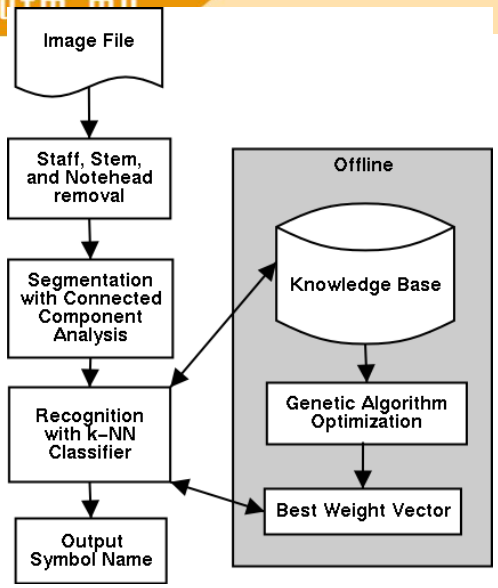
- DNA Computing
- Quantum Computing

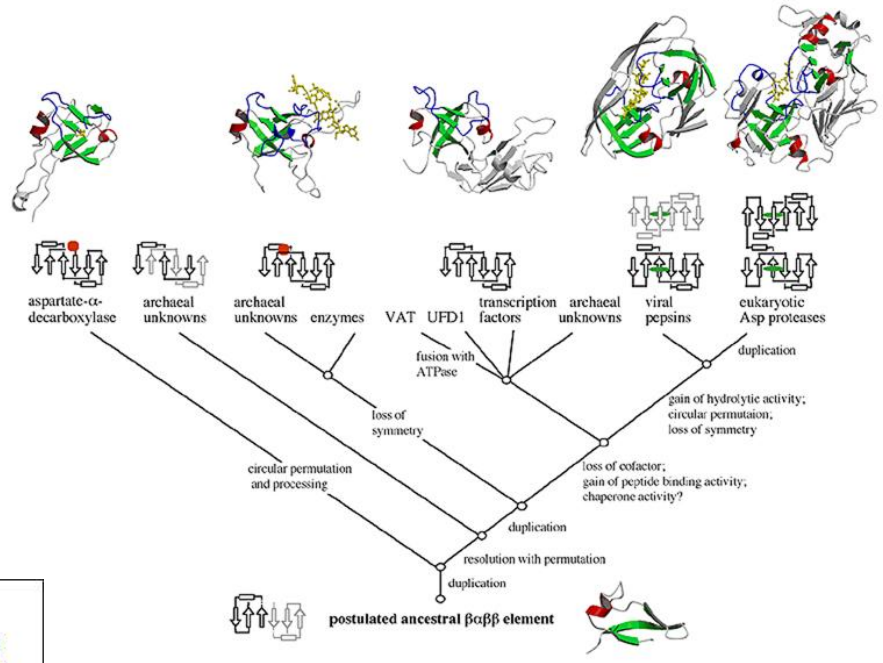


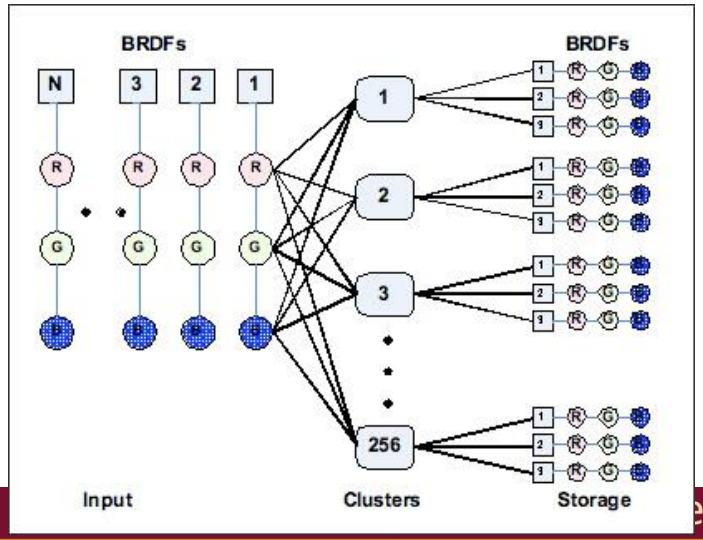
Importance of CI

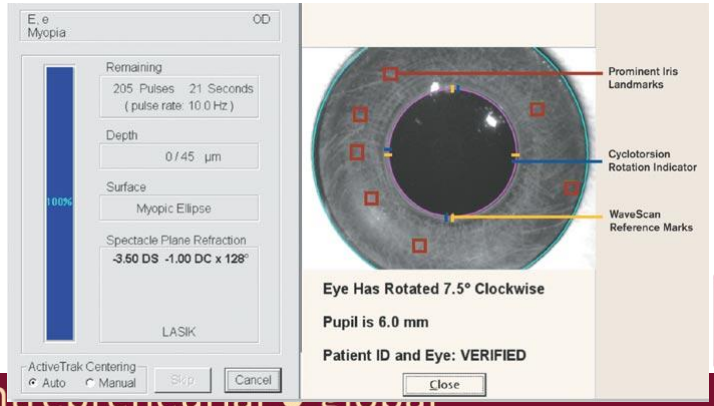
- Solve the problems in realistic situation (not ideal) such as fuzzy, imperfect or incomplete, only domain experts available.
- Complementary approaches with mathematical based algorithm solutions
- Encourage new technology development especially in Quantum and DNA computing.

Importance of CI





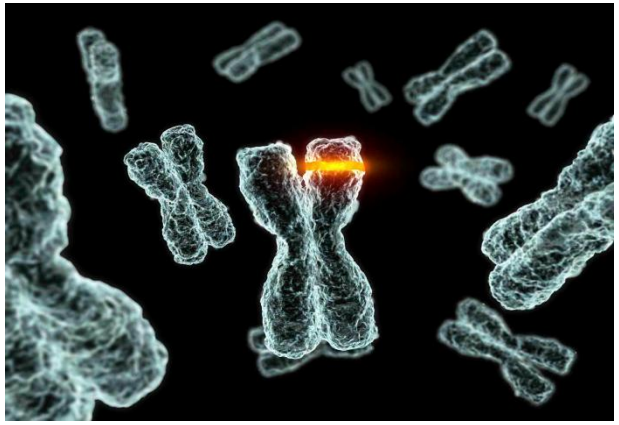
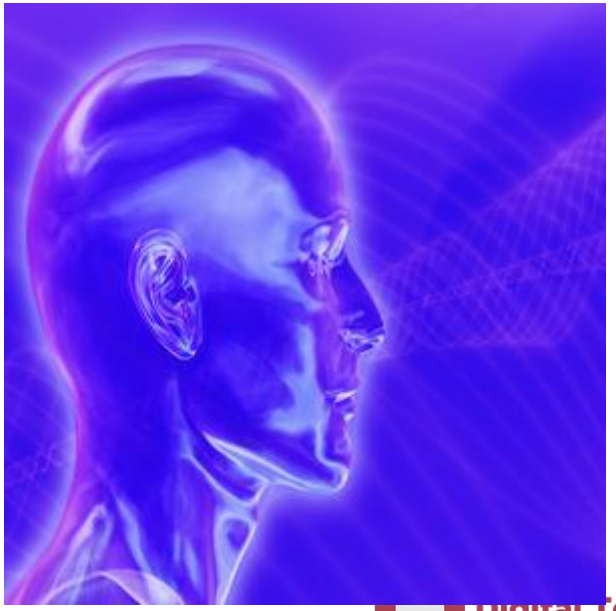




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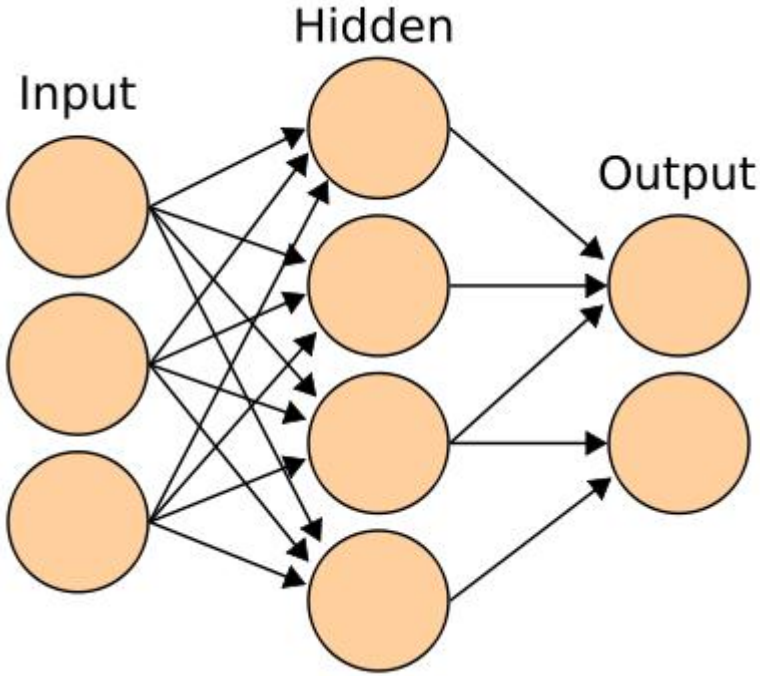
CI Techniques

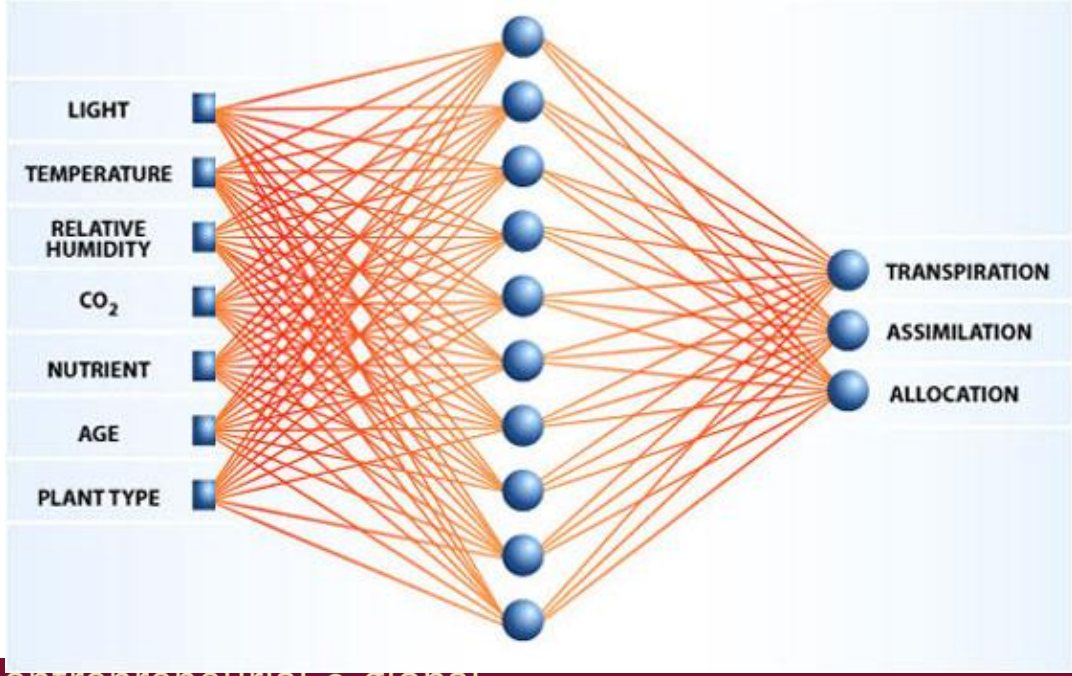
- Artificial Neural Network
- Genetic Algorithm
- Fuzzy Logic

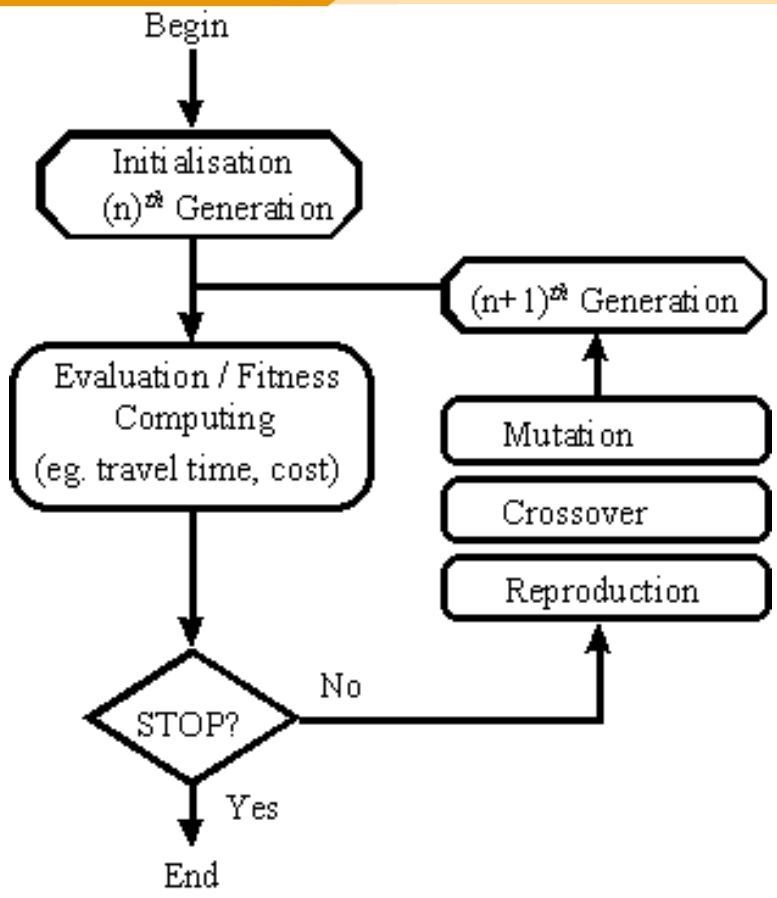
Artificial Neural Network







Genetic Algorithm



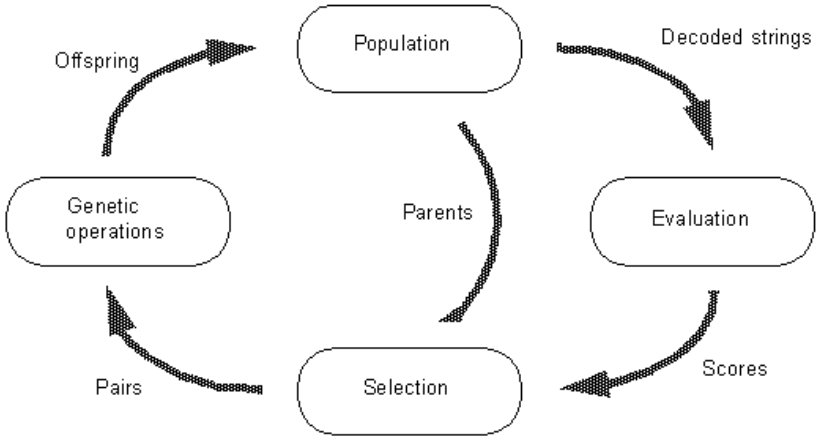
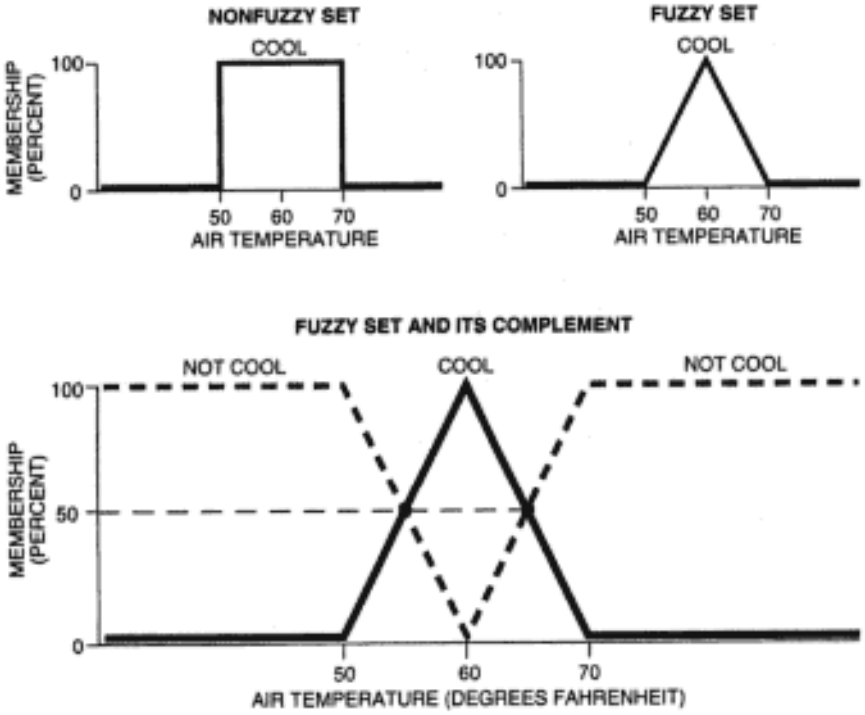
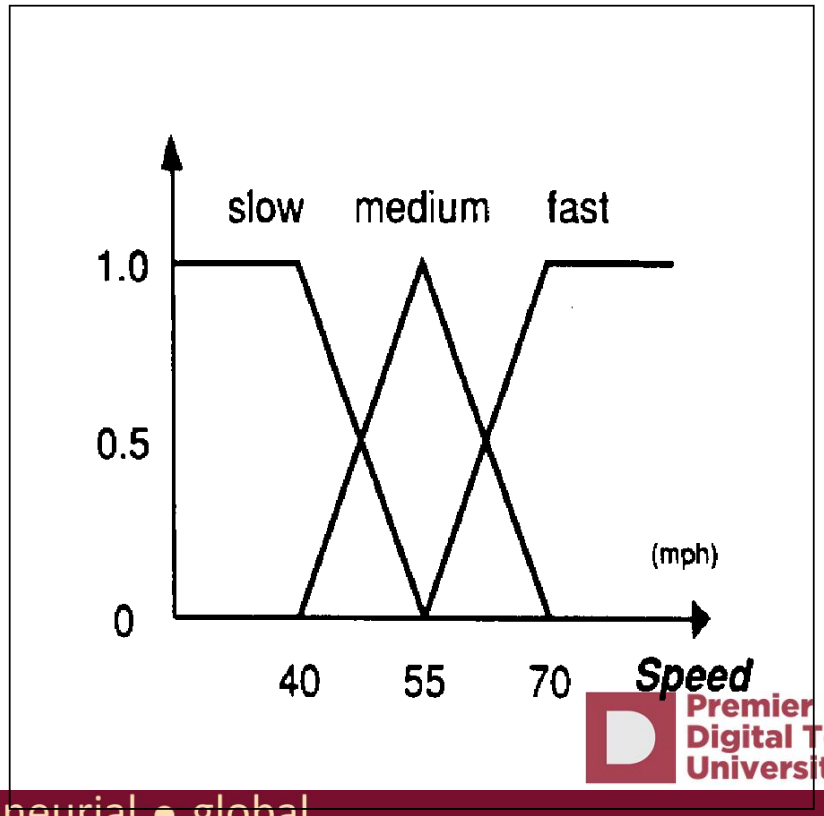


Figure 5.2: The "reproduction" cycle.

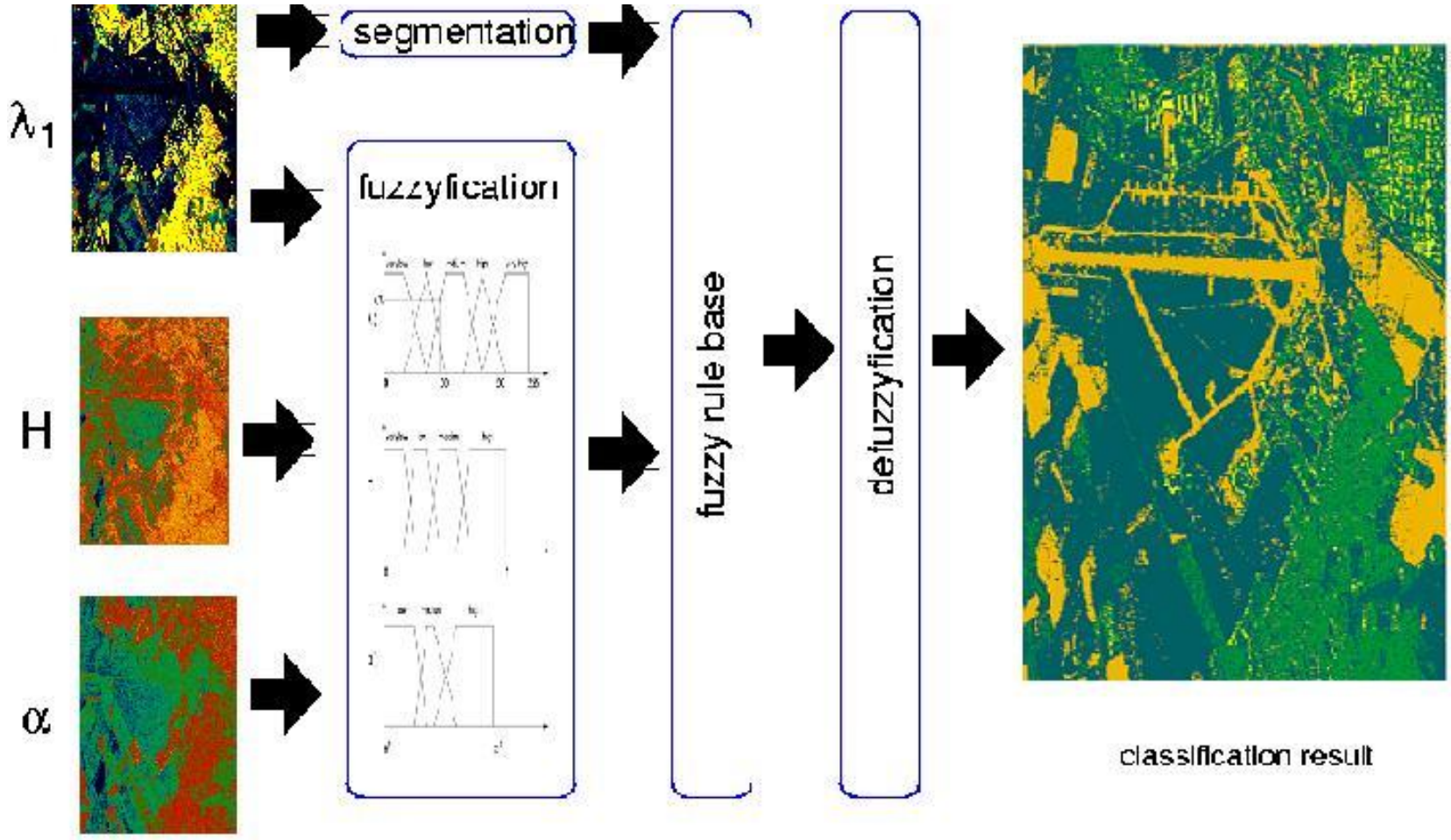


Fuzzy Logic


NONFUZZY SET
COOL
MEMBERSHIP (PERCENT)
100
0
50
60
70
AIR TEMPERATURE
FUZZY SET
COOL
100
0
50
60
70
AIR TEMPERATURE
FUZZY SET AND ITS COMPLEMENT
NOT COOL
COOL
NOT COOL
100
50
0
MEMBERSHIP (PERCENT)
50
60
70
AIR TEMPERATURE (DEGREES FAHRENHEIT)


slow
medium
fast
1.0
0.5
0
(mph)
40
55
70
Speed

Fuzzy Logic


λ1
H
α
segmentation
fuzzyfication
fuzzy rule base
defuzzyfication
classification result

Q & A

